Program 10: Class-based Component with API Fetching and Lifecycle Methods

10. Design a React application featuring a class-based component demonstrating lifecycle methods (componentDidMount, componentDidUpdate) to fetch and filter data from an external API with error handling.

Step-1: Create React Application

```
npx create-react-app data-fetcher
```

Step-2: Update App.js

```
import React, { Component } from "react";

const API = "https://jsonplaceholder.typicode.com/users";

class App extends Component {
  state = { d: [], f: [], s: "", l: 0, e: "" };

  componentDidMount() {
    this.get();
  }

  componentDidUpdate(p, prevState) {
    if (prevState.s !== this.state.s) {
      this.setState({
        f: this.state.d.filter(x =>
          x.name.toLowerCase().includes(this.state.s.toLowerCase())
        )
      });
    }
  }

  get = async () => {
    this.setState({ l: 1, e: "" });
    try {
      let r = await fetch(API);
      let d = await r.json();
      this.setState({ d, f: d, l: 0 });
    } catch {
      this.setState({ e: "Error fetching data", l: 0 });
    }
  }

  render() {
    let { f, s, l, e } = this.state;
    return (
      <div style={{ textAlign: 'center', padding: 20 }}>
        <h1>User Data</h1>
        {e && <p style={{ color: "red" }}>{e}</p>}
        <input
          placeholder="Search user..."
          value={s}
```

```
          onChange={x => this.setState({ s: x.target.value })}
          style={{ padding: 8, width: 250, marginBottom: 15 }}
        /><br />
        {l ? <p>Loading...</p> : (
          <table border="1" style={{ margin: '0 auto', borderCollapse: 'collapse', width: '80%' }}>
            <thead>
              <tr style={{ background: '#eee' }}><th>Name</th><th>Email</th><th>City</th></tr>
            </thead>
            <tbody>
              {f.length ? f.map(x =>
                <tr key={x.id}>
                  <td>{x.name}</td>
                  <td>{x.email}</td>
                  <td>{x.address.city}</td>
                </tr>
              ) : <tr><td colSpan="3">No results</td></tr>}
            </tbody>
          </table>
        )}
        <br />
        <button onClick={this.get} style={{ padding: '8px 16px' }}>Refresh</button>
      </div>
    );
  }
}

export default App;
```